

1. MODELS TRAINED / INTEGRATED SO FAR

This report separates models **trained by the VIVE project** from **pretrained models integrated and evaluated**. This distinction is important because AASIST, ECAPA, Silero VAD and IndicConformer are not project-trained from scratch.

Model/layer	Category	What has been done	Current role
Intent classifier	Project-trained	Trained using the project semantic/scam corpus and mapped to the VIVE intent taxonomy; evaluated with held-out data.	Live intent analysis.
Behavior classifier	Project-trained	Trained using the project semantic/scam corpus and mapped to the VIVE behavior taxonomy; evaluated with held-out data.	Live social-engineering behavior analysis.
Silero VAD	Pretrained	Integrated and tested as the speech-gating component.	Speech segmentation.
AASIST	Pretrained	Integrated; evaluated on ASVspoof data and VIVE target-domain samples.	Anti-spoof evidence.
ECAPA-TDNN	Pretrained	Integrated; embedding/reference workflow implemented; NO_REFERENCE handled explicitly.	Speaker consistency.
IndicConformer 600M	Pretrained	Integrated and benchmarked for Hindi/Tamil/Indic ASR; real device/backend streaming tests performed.	Speech-to-text.

2. PROJECT-TRAINED SEMANTIC MODELS

Intent classifier

- VIVE intent taxonomy includes NORMAL_CONVERSATION, OTP_REQUEST, PASSWORD_REQUEST, CARD_DETAILS_REQUEST, BANKING_CREDENTIAL_REQUEST, MONEY_TRANSFER_REQUEST, ACCOUNT_CHANGE_REQUEST, REMOTE_ACCESS_REQUEST, URGENT_ACTION, THREAT_OR_INTIMIDATION, CONFIDENTIAL_INFORMATION and UNKNOWN.
- The project records show a test macro-F1 of approximately 0.9219 on the evaluated supported classes, with validation macro-F1 around 0.9478.
- The earlier test coverage was uneven across labels; therefore per-class support remains important when reporting the result.

Behavior classifier

- VIVE behavior taxonomy includes AUTHORITY_IMPERSONATION, URGENCY, THREAT, FEAR, SECRECY, PRESSURE, REWARD_PROMISE and NORMAL.
- The project records show a test macro-F1 around 0.7095 when all eight labels were included, with lower values affected by zero-support/unsupported classes; supported-class performance was stronger.
- This model is used as a social-engineering behavior signal, not as a standalone fraud classifier.

3. ASR MODEL EVALUATION

ASR model	Evaluation / result recorded	Role
IndicConformer 600M	Hindi WER $\approx$ 0.1164; CER $\approx$ 0.0460. Tamil WER $\approx$ 0.2833; CER $\approx$ 0.1107. Real Tamil ASR was demonstrated.	Primary Indic ASR reference.
Whisper large-v3-turbo	Real 2-second processing was substantially slower in the benchmark; not selected as the primary mobile/streaming path.	Alternative evaluated.
Whisper tiny/base quantization experiments	On-device experiments compared float and int8 variants; the base model int8 experiment showed a useful size reduction with retained benchmark quality, while tiny int8 degraded more.	Mobile-ASR research/alternative path.

4. ANTI-SPOOF AND SPEAKER MODEL EVALUATION

Model	Recorded evaluation	Interpretation
AASIST	Proper ASVspoof 2019 LA evaluation produced EER $\approx$ 0.0133, close to the published benchmark reference. Three readouts agreed and an inverted control behaved as expected.	Implementation works on its benchmark domain. Target phone/call-like transfer requires separate validation.
AASIST on VIVE probe / phone-like audio	A VIVE probe showed chance-level/inconclusive discrimination; genuine device speech produced high spoof scores in some tests.	Do not treat raw AASIST output as universal synthetic-voice truth for the target domain.
ECAPA-TDNN	Pretrained speaker embedding model integrated. Without enrollment/reference, the system returns NO_REFERENCE.	Speaker evidence requires a legitimate reference protocol.

5. DATASETS / RESOURCES ACTUALLY USED OR EVALUATED IN THE BUILD

Dataset/resource	Use recorded in project	Category
ASVspoof 2019 LA	AASIST implementation/evaluation and benchmark validation.	Anti-spoof evaluation
ASVspoof 2021 / public anti-spoof resources	Referenced/used for anti-spoof component testing according to project test documentation.	Anti-spoof evaluation
FLEURS — Hindi/Tamil	ASR component testing and multilingual speech evaluation.	ASR
LibriSpeech	Clean speech testing for speaker/ASR-related experiments.	Speech/speaker evaluation
VIVE semantic/scam corpus	Training/evaluation of intent and behavior classifiers; project records describe approximately 85,602 records.	Intent/behavior training
VIVE manual test dataset	Small controlled test utterances for live/manual functional checks; not used to claim accuracy.	Functional testing

6. DATASET COUNT — HOW TO STATE IT

For reporting purposes, separate the counts by role rather than presenting one misleading total:

- **1 main project semantic corpus** used for the project-trained intent/behavior classifiers (approximately 85,602 records in the project records).
- **Multiple public speech evaluation resources** were used/referenced across anti-spoof, ASR and speaker testing, including ASVspoof, FLEURS and LibriSpeech.
- **1 small VIVE manual test set** was prepared for functional/live testing.
- The project documentation also identifies additional datasets for planned expansion; those should be counted separately from datasets already used.

7. DATASETS PLANNED FOR FURTHER EXPANSION

- ASVspooF 5 / permitted ASVspooF releases for broader generator-disjoint anti-spoof evaluation.
- IndicSynth for Indian synthetic-speech evaluation.
- Vaani and other legally permitted Indian speech resources.
- VoxCeleb1/2 for speaker foundation/evaluation.
- ScamShield and Hinglish/English financial scam text for semantic expansion after license/provenance review.
- Consented/project-generated samples for human-scam and synthetic-legitimate scenarios.

8. CURRENT MODEL PIPELINE

Audio → Silero VAD → AASIST + ECAPA → IndicConformer ASR → Intent + Behavior → Context → OOD/uncertainty → Risk fusion → Temporal risk → Policy.

9. REPORTING RULE

AASIST/ECAPA/IndicConformer are **pretrained integrated models**; intent and behavior are the main **project-trained classifiers**. Risk/OOD/temporal logic is a project-level intelligence layer. Performance numbers should always be reported with the dataset, split and evaluation protocol that produced them.